Document continuity — logic as implemented

Which document each page belongs to, and the values a document's pages share. What the code does today (the weighted revamp, 2026-10-18). The design, the measurement behind the weights and the encounter-type half still to build: `CONTINUITY_ENCOUNTER_REVAMP.md`.

Engine	`core-pipeline/stages/lib/continuity/engine.py` (`assign`, `_gate`, `_weighted`, `document_finals`)
Signals	`core-pipeline/stages/lib/continuity/signals.py`
Stage	`continuity`, seq 87 — after `page_subtype` and `dos_extract`, before `imaging_final`
Settings	`stages/lib/keyword-canon/continuity_canon.json` (version 3, reloads on change)
Writes	`page_continuity_results` (incl. `score`, `review_required`), `imaging/<chart>_continuity.csv`
Final values	written by `imaging_final` (seq 100) into `imaging_final` / `<chart>_final.csv`
Database changes	`schema/changes_2026-10-18.sql` §4 (`decided_by` gains `weighted`)

1. The question

For every page: **does it continue the previous page's document, start a new one, or is it unclear?** The pages of one document then share the first page's **Final** page type and **Final** DOS. Each page keeps its own Extracted values.

Getting it wrong is asymmetric: merging two encounters is a clinical error that reaches coding; splitting one is a nuisance a reviewer catches. Unclear pages are therefore **flagged**, not guessed.

2. Inputs per page

Input	From
Text	the page's OCR text (Final 2 → Final 1 → Tesseract)
Line positions	Final 2 JSON line polygons, else Final 1 words grouped into lines, else plain text lines
Section headers	the OCR JSON (`section_headers`, matched to `section_header_canon.json`)
Own DOS (From, To)	`dos_extraction_results` — the Extracted DOS
Page type (L1), sub-type, full path, confidence	`page_classification` (axes)
Title on the page	page classification (`axes.title_of`, else `keyword_title_hit`)
Printed page number "N of M"	`additional_page_details` (key/value stage), else read from the header / footer bands
Signed	`provider_signature.csv` `signature_present = y`, or a signature block in the text

Input	From
Signed in tail	a signature block in the page's **last third** of lines
Skipped	blank / junk / duplicate (final verdict)

Bands: header = a line whose top is above **10%** of the page height; footer = a line whose bottom is below **92%**. Without positions: the first 4 and last 3 lines.

---

3. The decision

3.1 Skipped pages

Blank, junk and duplicate pages belong to no document. A **blank or junk** page also **ends** the open document (a physical separator); a **duplicate** does not (a scanning artefact) and takes its original's document and values.

3.2 Gates — the first that applies decides

#	Gate	Result	`decided_by`
1	First page of the chart	new	`first_page`
2	A blank or junk page came just before	new	`blank_junk`
3	Own DOS **equal** to the open document's (From and To)	**continue**	`same_dos`
4	Own DOS, **different**	**new**	`new_dos`
5	Printed page numbers follow on: "3 of 5" → "4 of 5"	**continue**	`pagination`
6	The previous page was printed "page N of N"	new	`page_number_end`

DOS outranks pagination: a different date is a different encounter, whatever the footer says. **Pagination never starts a document**: it confirms a continue (5) or closes one (6) — a misread digit must not split a document.

3.3 Weighted evidence — an undated page no gate decided

```
score = Σ weight × value        every value in [0, 1]

score >= +4     continue         confidence medium   decided_by weighted
score <= −4     new document     confidence medium   decided_by weighted
otherwise       unknown          confidence low      review_required, stays in the open
document
```

Signal	Weight	Value	Status
`section_forward`	**+10.0**	the next page's sections follow the previous page's in note order (§4.3), 0..1	fitted
`page_type_change`	**−7.0**	(1 − same-type probability) × confidence in the new type (§4.4)	estimated
`cur_has_title`	**−6.0**	the page carries a title for its page type	estimated
`same_visit_number`	+5.4	both pages print the same visit / encounter / FIN / CSN number	fitted
`page_fill`	+5.1	how far down the previous page's body text reaches (§4.2)	fitted
`header_line_repeats`	+4.7	a header line (≥ 12 characters) printed on both pages	fitted
`prev_signed`	−4.4	the previous page is signed anywhere	fitted

Signal	Weight	Value	Status
`body_cosine_char`	+3.7	character tf-idf similarity of the previous page's tail and this page's head (§4.1)	fitted
`full_path_same`	+3.0	the same page type at every level (L1 / L2 / L3)	estimated
`footer_line_repeats`	+2.8	a footer line printed on both pages	fitted
`body_cosine_word`	+1.0	word tf-idf similarity of the same blocks	fitted
`cur_opens_note`	+0.5	the page opens on Chief Complaint / HPI	fitted
`prev_signed_in_tail`	−0.4	the previous page's signature is in its tail	fitted

`fitted` = measured on 170 labelled pairs from 15 charts (balanced accuracy 0.79 ± 0.19, split by chart); `estimated` = specified after the measurement, to fit before trusting. The `measurement` block in the canon carries the provenance.

3.4 The abstain band

A page scoring between −4 and +4 is **unknown**: it stays in the open document (leaving a page in place is the recoverable error; starting a spurious document fragments an encounter) and is flagged `review_required`. Never widen the band to lower the abstain rate — those are the pages most likely to be wrong.

3.5 Signature terminator

A previous page with a **signature block in its tail** closes its document even when the score kept it open (`decided_by signature`). Gates 3 and 5 (same DOS, page numbers) still win, because they decide first.

---

4. The signals in detail

4.1 Text running from one page into the next

1. **Strip the furniture**: lines of ≥ 12 characters printed on both pages (patient banner, footer template) are removed — they appear on every page of the chart and would swamp the body. (They are what `header_line_repeats` / `footer_line_repeats` score.)
2. **Tail and head blocks**: the previous page's last body lines, walking back in groups of three lines until 20 words; the next page's first body lines, forward, likewise.
3. **TF-IDF cosine** on those blocks — character 3–5 grams within words, and word 1–2 grams; `min_df` 2, sublinear tf; fitted on **every block of the chart** (never per pair). Normalised: `min(1, cosine / 0.25)` for characters, `/ 0.15` for words.

4.2 Page fill

Where the previous page's body ends: the bottom of its last line **starting above the footer band**, divided by the footer top (0.92), clipped to 1.

Fill	Value
≤ 0.60	0 — the document probably ended there
0.60 – 0.92	graded: (fill − 0.60) / 0.32
≥ 0.92	1 — the text runs on

Without line positions: 1 when the page has ≥ 28 lines or ≥ 1,800 characters, else 0; when that text proxy swings a weighted decision, the page is flagged for review.

4.3 Section order

Note order (`section_order` in the canon, each step mapped to `section_header_canon.json` names): chief complaint → HPI → review of systems → past history → social → family → medications → allergies → vitals → physical exam → labs → imaging → problem list → assessment / diagnosis → plan → patient instructions → follow-up.

`section_forward = clip((first section of this page − last section of the previous) / steps × 4, 0, 1)`. Moving backwards counts 0 (no evidence either way).

4.4 Page type change

Previous	This page	Confidence	Same-type probability	Value	× −7
Progress Note	Lab	90%	10%	0.81	−5.7
Progress Note	Lab	60%	40%	0.36	−2.5
Progress Note	Progress Note	95%	95%	0.05	−0.3

5. Review flags (`review_required`)

Flag	When
Unknown	the score fell inside the band (§3.4)
Long run without page numbers	a document of more than **25** pages where no page prints a page number (a footer template may be binding unrelated documents) — every page of it
Text-proxy fill	page fill came from the text proxy (no line positions) and decided

6. Links and position

Field	Meaning
`link`	`strong` — joined by same DOS or page numbers; `weak` — by the weighted score; `unknown` — kept across the abstain band
`link_strength`	`strong` while every link back to the first page is strong, else `weak`
`position`	`single` / `first` / `continue` / `last` (review-ui: "Continue (Doc 3)")
`start_confirmed`	the first page has a title, or prints "page 1"
`score`	the weighted score (weighted / signature decisions only)
`evidence`	e.g. "same DOS 2024-03-14", "no DOS of its own; page 3->4 of 5", "score +10.4 (page_fill +5.1, full_path_same +3.0, footer_line_repeats +2.8, page_type_change -1.1)", "score +0.3 between -4 and +4: kept in the document, for review (…)"

7. What a document shares — the Final values

Unchanged by the revamp (`document_finals`):

Final value	Rule
Page type	the first page's (each page keeps its own sub-type)
DOS	the first page's, on every page
DOS of a stand-alone non-visit document	every page a `default_final_dos` type (Patient Demographics, Patient Information, Patient Registration, Growth chart, Problem List, Immunizations) → **2222-02-02**; such a page inside a visit takes the visit's DOS
Encounter type	the first page's

8. Settings (`continuity_canon.json`, version 3)

Block	Holds
`pagination`, `bands`, `default_final_dos`	as before
`weights`	each signal's `w`, measured `lift`, `coverage`, `status` (fitted / estimated)
`decision`	`continue_at` 4.0, `new_at` −4.0
`semantic`	furniture 12 chars, 20-word blocks, char 3–5 / word 1–2 grams, `min_df` 2, divisors 0.25 / 0.15
`page_fill`	`ended_at` 0.60, `continues_at` 0.92, graded, text proxy 28 lines / 1,800 chars
`section_order`	the steps and the section names that count as each; `opens_note`
`signature_terminator`	on; tail = the last 33% of lines
`review_triggers`	unknown, long run 25, text-proxy fill
`measurement`	provenance of the weights
`document_end`	kept for reference; the ladder that read it is gone

The loader refuses a file whose `weights` name a signal the engine does not produce, or whose band is inverted (`new_at ≥ continue_at`).

9. Worked example

Page	Own DOS	Evidence	Decision	Doc
1	2024-03-28	—	first page	1
2	—	"page 2 of 3" after "page 1 of 3"	continue (`pagination`)	1
3	—	banner + footer repeat, text to the bottom, sections HPI → Plan	continue (`weighted`, +27.6)	1
4	—	previous page ends "Electronically signed by …"	new (`signature`)	2
5	—	Lab after Progress Note at 90%, a Lab title, text ended at 40%	new (`weighted`, −11.7)	3
6	—	text to the bottom (+5.1), but Lab after Lab at 60% with a title	unknown, flagged (`weighted`, −0.9)	3

Page	Own DOS	Evidence	Decision	Doc
7	2024-06-03	—	new (`new_dos`)	4
8	blank	no ink	skipped; ends Doc 4	—